Manuel Palenque — Keynote Speaker Profile & Interactive Installations Dossier (2026/2027)

> **Bridging Fortune 500 Creative AI & Immersive Systems, Global Theatrical Spectacles, and Avant-Garde Interactive Public Art.**

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1. Executive Speaker & Curatorial Profile

Manuel Palenque (b. 1986, Buenos Aires) occupies a rare, high-authority intersection in the global creative technology landscape: he executes at the highest enterprise tier while remaining deeply embedded in the experimental, hardware-hacking underground of Latin America.

Over a 15+ year career across **Tokyo, Hong Kong, Berlin, Venice, Barcelona, Buenos Aires, and the United States**, Palenque has built his practice across three complementary pillars:

1. **Enterprise Creative Technology & AI Leadership (Superside):** As **Senior Creative Technologist (Immersive & Creative AI Architectures)** at global design unicorn **Superside**, he architects scalable AI and immersive production workflows for Fortune 500 clients including **Google, Meta, and Reddit**. He is the author of the *Immersive Design & Creative AI Architectures* methodology (superside-immersive.github.io/course), training creative teams to transition from manual software execution to dynamic system orchestration.
2. **Global Entertainment Spectacles & Monumental Public Art:** As former **Technical Video Director for Fuerza Bruta** (Tokyo / Buenos Aires), Lead A/V & Sensor Synchronization Director for Argentina's **Fiesta Nacional de la Vendimia** (500+ stage performers), VR Experience Lead for **Cirque du Soleil's Messi10 Challenge**, Global Live Production Lead for **Bright Moments** (Venice Biennale context, Paris, London, Buenos Aires), and custom LED control engineer for the 45-meter **Coloso de Tecnópolis** with DOMA Collective, he has engineered mission-critical real-time systems experienced by hundreds of thousands of people.
3. **Independent Curation & Grassroots Media Lab Leadership (DataRaíz / Ciclo Negro):** He is the founder of **DataRaíz** (dataraiz.io), Buenos Aires' independent laboratory for art, science, and technology, and creator of **Ciclo Negro** (ciclonegro.dataraiz.io), a 12-hour interactive art and live audiovisual festival at **ArtHaus Central**.

Target Booking & Commissioning Contexts

- **Technology, Design & Culture Conferences:** SXSW, Cannes Lions, MUTEK Forum, AWE (Augmented World Expo), OFFF, ODSC AI, EAI ArtsIT, SIGGRAPH, Ars Electronica.
- **Speaker Bureaus:** The Lavin Agency, Chartwell Speakers, CAA / UTA Innovation Speakers.
- **Curatorial Agencies, Public Art & Brand Commissions:** MTArt Agency, RADAR Curatorial, StandardVision, Times Square Arts (Midnight Moment), Cultural Counsel, Superblue, Mercer Labs.
- **Residencies, Grants & Biennials:** Code:ART (Palo Alto), Bemis Center for Contemporary Arts, Pioneer Works, CURRENTS New Media, Eyebeam, Knight Foundation.

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2. Three Complete Keynote & Lecture Abstracts (2026/2027)

Keynote 1: *Syncro A/V — Designing Tightly Coupled Real-Time Audiovisual & Lighting Systems*

- **Format:** 45–60 min Illustrated Keynote or 90-min Technical Lecture-Demo (with live real-time shader/audio/DMX examples)
- **Reference Portal:** https://mpalenque.github.io/syncro-av/

ABSTRACT (200 WORDS)

Why do certain live performances feel like a single, physical organism while others feel like a concert playing in front of a screensaver? In *Syncro A/V*, creative technologist and live audiovisual artist Manuel Palenque unpacks the perceptual neuroscience and real-time systems engineering behind tightly coupled sound, image, and spatial lighting. Drawing from 15 years of building high-stakes shows—from directing technical video for *Fuerza Bruta* and synchronizing 500 performers at *Vendimia* to performing at *MUTEK* and Tokyo's *SuperDeluxe*—Palenque demonstrates how seven automatic mechanisms of human multisensory integration (*synchresis*, *the ventriloquism effect*, *the McGurk effect*, *the temporal binding window*, *common fate*, *rhythmic entrainment*, and *prediction/reward*) govern every audience in a dark room. Moving from perceptual theory to live production architecture, the session reveals how to replace pre-rendered timelines with bi-directional state machines linking electronic sound synthesis, GPU particle physics, custom shaders, and physical DMX/Art-Net lighting and lasers. Attendees learn how understanding *why* sound and image fuse in the brain transforms every technical synchronization parameter into a powerful compositional and dramaturgical decision—including when to maintain sub-millisecond lock and when to let entropy tear the system apart.

KEY TAKEAWAYS

1. **Perceptual Composition:** How to apply the seven principles of multisensory integration (*synchresis*, *temporal binding window*, *rhythmic entrainment*) to design visceral audiovisual fusion.
2. **Unified Signal Architecture:** How to architect low-latency, bi-directional pipelines connecting audio feature extraction (FFT, transients, sub-bass pressure) directly to GPU shaders (`vvvv` / `TouchDesigner` / `WebGPU`) and physical DMX/laser hardware.
3. **Dramaturgy of Entropy:** How to balance algorithmic precision with controlled systemic failure so live performances remain unpredictable and alive.

TARGET AUDIENCE

Creative technologists, audiovisual artists, live show directors, lighting and stage designers, electronic musicians, experiential agency leads, and media art curators (MUTEK Forum, SXSW Music/Tech, OFFF, university media labs).

Keynote 2: *From Stage to WebGPU — Building Scalable Immersive Experiences and 3D Digital Twins for Live Performance*

- **Format:** 45–60 min Technical & Production Case-Study Keynote
- **Reference Portals:** https://mpalenque.github.io/ · https://mpalenque.github.io/curso/

ABSTRACT (200 WORDS)

In large-scale live entertainment, museum commissions, and international festival touring, the biggest enemy of artistic ambition is time: production teams frequently face complex multi-screen, projection-mapped, or DMX-heavy stages with less than an hour of on-site rehearsal before doors open. In this case-study-driven keynote, Manuel Palenque reveals how **3D Stage Digital Twins**, **LiDAR spatial scanning**, and browser-native **WebGPU / WebSockets** architectures are transforming how immersive experiences are designed, rehearsed, and scaled. Using real-world post-mortems—including building a full virtual 3D stage and DMX simulation to create his monumental live A/V piece *COLAPSO* at Tecnópolis in 30 days with zero venue rehearsal; utilizing remote LiDAR

scans to calibrate fluid simulations and physical water reflections inside a Renaissance church for *Bright Moments Venezia*; staging an un-rehearsed 8-hour, 200-piece live generative marathon in Buenos Aires; and deploying a zero-download QR/WebSockets system that turned the *Buenos Aires Planetarium* into a collective 360° instrument—Palenque provides a battle-tested blueprint for de-risking ambitious spatial productions. Attendees will discover how virtual pre-visualization and distributed web architectures allow small, agile teams to deliver stadium- and museum-grade immersive works on tight budgets and timelines.

KEY TAKEAWAYS

1. **Zero-Rehearsal Production via Digital Twins:** How to build accurate 3D stage digital twins (screens, sightlines, virtual DMX universes, and LiDAR point clouds) to program and stress-test shows before load-in.
2. **Frictionless Crowd Participation at Scale:** How to engineer zero-download WebSockets and WebGPU mobile controllers that turn ratusan/thousands of passive spectators into co-creators of a shared 360° projection without Wi-Fi collapse.
3. **Cross-Platform Scalability:** Designing modular shader and lighting pipelines that adapt seamlessly from browser previews to 1,200-nit LED walls and fulldome planetariums.

TARGET AUDIENCE

Experiential producers, technical directors, XR/WebGPU engineers, museum and biennial curators, architectural projection specialists, and festival production managers (AWE, SIGGRAPH, SXSW, ISE, LDI).

Keynote 3: *Creative Coding & AI Transitions — New Workflows for Interactive Installations and Audiovisual Art*

(Alternate Conference Title for Cannes Lions / SXSW: ***Democratizing Creative AI from LATAM to Global Scale — From Software Users to System Architects****)*

- **Format:** 45–60 min Strategic & Creative Keynote with Live Workflow Demos
- **Reference Portals:** https://superside-immersive.github.io/course/index.html · https://superside-immersive.github.io/course/talk.html

ABSTRACT (200 WORDS)

Artificial Intelligence often feels like a closed black box built by and for the Global North—promising effortless automation while producing homogenized, soulless output. Yet in Latin America, where resource scarcity and "hacking limits" are foundational cultural methodologies, AI and creative coding are being weaponized for radical artistic agency. Presenting a unique dual perspective—as Senior Creative Technologist at global unicorn **Superside** (building enterprise AI and immersive workflows for Google, Meta, and Reddit) and as founder of the Buenos Aires media lab **DataRaíz**—Manuel Palenque outlines a paradigm shift from **technical execution** to **dynamic system orchestration**. Rejecting both blind techno-utopianism and creative paralysis, Palenque demonstrates how individual artists and lean design studios can use AI coding agents with "critical distance" to build custom real-time pipelines across **TouchDesigner, Python, vvvv, custom GLSL shaders, and WebXR**. Through live demonstrations and global case studies, this talk shows how adding strict artistic constraints, human taste, and physical-world friction allows independent creators to subvert corporate-grade tools—transforming scarcity into a competitive methodology and empowering creatives to build their own instruments rather than merelyrenting someone else's software.

KEY TAKEAWAYS

1. **From Software Users to System Architects:** A practical methodology for using AI coding agents to bridge TouchDesigner, Python, shaders, and WebXR without decades of traditional software engineering.

2. **Constraints & Friction as Creative Engines:** Why AI only becomes artistically compelling when subjected to strict boundaries, physical sensors, real-time performance risk, and human curation.
3. **The Global South Playbook:** How to turn resource constraints into high-velocity innovation, redirecting enterprise platforms and workflows to sustain independent art communities like *DataRaíz* and *Ciclo Negro*.

TARGET AUDIENCE

Creative directors, innovation leads, brand executives, independent artists, designers, educators, and cultural policymakers (Cannes Lions, SXSW, ODSC AI, OFFF, design schools).

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3. Interactive Installations & Commissions Dossier

All works documented below are real, battle-tested productions with complete technical specifications, video documentation, and modular hardware/software architectures ready for **museum exhibition, public art commissioning, biennial presentation, or brand activation**.

3.1 `COLAPSO` & `COLAPSO 3D Digital Twin` (2022–2026)

- **Category:** Monumental Audiovisual & Reactive Lighting Installation / Live A/V Performance + 3D Stage Digital Twin
- **Premiere & Venues:** Tecnópolis (Buenos Aires, 2022); ArtHaus / Ciclo Negro (Buenos Aires, 2025/2026)
- **Tech Stack:** `vvvv`, GPU real-time particle systems, custom GLSL/HLSL shaders, `DMX512 / Art-Net` reactive stage lighting, custom 3D Stage Digital Twin simulation software, stereo sub-bass spatial audio.
- **Curatorial Concept:** *COLAPSO* explores the unavoidable progression of entropy and the fragility of human-imposed order. In both its installation and live performance modes, real-time sound and generative GPU particle systems communicate bi-directionally: acoustic energy shapes the visual landscape until the visual and lighting systems progressively destabilize, overpowering harmonic balance and plunging the space into high-contrast digital disintegration.
- **The 3D Digital Twin Innovation:** Conceived under an extreme 30-day production sprint for Tecnópolis with virtually zero on-site rehearsal time, Palenque engineered a complete **3D Digital Twin** of the venue's screen geometry, architectural sightlines, and DMX lighting fixtures. Every camera angle, luminance interaction, and DMX cue was programmed and rehearsed in virtual space prior to physical load-in.
- **Technical & Spatial Requirements:**
  - **Display:** Fully scalable to custom LED walls (minimum **1,200 nits**), large-scale architectural projection mapping (minimum **300 nits** surface luminance, **≥ 10,000:1** native contrast ratio), or multi-surface configurations.
  - **Environment:** Strict ambient light control (**total or near-total darkness** mandatory to preserve absolute deep blacks) + atmospheric hazer.
  - **Audio:** Full-range concert PA with clean, high-headroom sub-bass extension (2x balanced XLR/DI stereo inputs).
- **Documentation Links:**
  - **Video:** https://youtu.be/UEpg5IswDS8
  - **Photo Album:** https://photos.app.goo.gl/2YX1uSQSutED1mxHA
  - **Stage & Digital Twin Deck:** https://xn--kyla-nauvis-zbb.my.canva.site/colapso-proposal-performance

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3.2 `DataRaíz: Ciclo Negro` (2025–2026)

- **Category:** 12-Hour Interactive & Audiovisual Art Festival, Curatorial Platform & Turnkey Institutional Takeover
- **Venue:** ArtHaus Central (Sala Oval / Piso 7 & Multi-Space), Buenos Aires, Argentina
- **Role:** Founder, Curator, Technical Director & Live A/V Artist
- **Production Scale & Specs:** 12-hour continuous programming integrating interactive installations, hands-on technical masterclasses (*Syncro A/V*), critical panels on art and technology, and nocturnal live A/V performances. Engineered around a modular indoor P2 LED wall (4m × 3m), multi-source video processing (`vMix` / `vvvv` / `TouchDesigner`), synchronized DMX stage lighting, and quad/stereo sound reinforcement.
- **Curatorial Concept:** Created through **DataRaíz** (`dataraiz.io`), *Ciclo Negro* is a curatorial platform and temporary autonomous zone where advanced technology ceases to be a corporate utility and becomes a sensitive, collective language. It bridges emerging experimental coders and established audiovisual artists in an unfinished, laboratory-like public format.
- **Touring / Institutional Takeover Readiness:** Available as a curated "Pop-Up Showcase & Symposium" for international partner institutions (e.g., Pioneer Works, Gray Area, CultureHub, MUTEK satellite programs), bringing a curated roster of Latin American real-time media art alongside workshops and live A/V sets.
- **Documentation Links:**
  - **Festival Portal:** https://ciclonegro.dataraiz.io/
  - **DataRaíz Media Lab:** https://dataraiz.io
  - **What is DataRaíz (Video):** https://youtu.be/kmPwVNuiHjI

3.3 `FullDome 360° Interactive System` (2023–2026)

- **Category:** Participatory 360° Fulldome Projection & Multi-User WebSockets Architecture
- **Premiere & Context:** Buenos Aires Planetarium / Fulldome 360° Venues (2023)
- **Tech Stack:** Custom real-time 360° domemaster rendering engine (`vvvv` / `TouchDesigner` / custom shaders), distributed low-latency **WebSockets** server, zero-download mobile **WebApp** accessed via dynamic on-dome **QR Code** projection.
- **Curatorial Concept:** Traditional planetarium and immersive dome shows relegate the audience to passive spectatorship. *FullDome 360° Interactive System* transforms the entire dome into a **shared, collective visual instrument**. Upon scanning a projected QR code with their smartphones—with zero app download or registration friction—hundreds of spectators simultaneously inject real-time trajectories, forces, and parameter shifts into the 360° generative visual field overhead.
- **Technical Innovation:** Specifically engineered to overcome high-density crowd network congestion, the WebSockets state-synchronization layer operates with sub-frame responsiveness even under degraded or offline local Wi-Fi conditions, aggregating hundreds of individual micro-interactions into a coherent, high-resolution 360° visual narrative.
- **Commissioning & Adaptation:** Natively adaptable to planetariums, geodesic festival domes, immersive projection rooms (360° wall/floor rooms), and large-scale public/stadium LED facades seeking instant crowd participation.
- **Documentation Links:**
  - **Video Preview (Canva / Reel):** https://www.canva.com/design/DAHA9LOF4dE/9Losr1ymieky10R52SWIDg/edit
  - **Portfolio Entry:** https://mpalenque.github.io/

3.4 `Bosque` & `Techo` — Verano Tecnópolis / Electronic Arts Laboratory (2023)

- **Category:** Large-Scale Multi-User Interactive Laser, Projection, Fluid & Reactive DMX Environments

- **Venue:** Tecnópolis (Feria IT Joven — Laboratorio de Artes Electrónicas, invited by Plan), Buenos Aires (2023)
- **Tech Stack:** vvvv, **Microsoft Kinect / Depth Sensor Arrays**, real-time 2D/3D GPU fluid simulations, synchronized **DMX512 / Art-Net moving-head beams**, **ILDA/DMX lasers**, large-format projection mapping, and 3D spatial pre-visualization models.
- **Curatorial & Technical Breakdown:**
    - **`Techo` (Interactive Ceiling Fluid & Moving-Light Matrix):** An overhead architectural installation synchronizing a matrix of robotic moving-head light beams with a real-time digital fluid simulation projected across the ceiling structure. Every angular sweep and intensity variation of the physical light beams acts as a physical force vector inside the digital fluid solver, while Kinect depth sensors track visitors below—creating a seamless perceptual fusion where physical light beams appear to stir and Agitate liquid digital matter floating in mid-air.
    - **`Bosque` (Site-Specific Interactive Natural-Digital Ecosystem):** Installed within a wooded/organic environment at Tecnópolis, *Bosque* uses Kinect motion capture, volumetric projections, and lasers to suspend responsive digital entities in mid-air without visible screens. As visitors walk through the space, the floating generative organisms react to human proximity and gesture, questioning the imposition of synthetic visual technology onto natural ecosystems.
- **Documentation Links:**
    - **Techo Real-Time Simulation & Installation Video 1:** https://www.youtube.com/watch?v=TXt8exDiNhA
    - **Techo Installation Video 2:** https://www.youtube.com/watch?v=2K_x0nrlNG0
    - **Techo Backstage Video 3:** https://www.youtube.com/watch?v=xaYuqlP2Flk

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3.5 `Wall — Adidas NMD` (2022)

- **Category:** Commercial Interactive LED Wall & Kinetic Depth-Tracking Installation
- **Client & Venue:** Commissioned by **Adidas** (Adidas NMD Campaign Activation, Buenos Aires, 2022)
- **Tech Stack:** vvvv, **Kinect Depth Sensor Array**, real-time GPU Navier-Stokes fluid dynamics, custom reactive shaders, large-format high-brightness indoor LED wall.
- **Curatorial & Experiential Concept:** Designed to capture attention in high-foot-traffic retail and cultural environments *before* a passerby consciously decides to stop, *Wall* turns an architectural LED facade into a living, perceptual membrane. Depth sensors track the velocity, silhouette, and trajectory of every person walking past, translating human movement into organic, high-contrast fluid wakes and generative typographic/geometric distortions in real time. Each visitor leaves a unique, ephemeral visual signature that collides and blends with the trails of others.
- **Commercial Commissioning Value:** Demonstrates turnkey capability to deliver ultra-reliable, zero-latency, unattended interactive brand installations for flagship retail, trade shows, and public lobbies.
- **Documentation Links:**
    - **Portfolio Video (`#adidas-nmd`):** https://mpalenque.github.io/#adidas-nmd
    - **Project Photo Archive:** Google Photos Album

3.6 Physical-Digital, Shader & Physics R&D Works (`Fluids`, `Colisión`, `Rope`, `Agua`)

A modular suite of real-time physical simulations and hybrid material studies adaptable to gallery monitors, architectural LED totems, site-specific sculptures, and Times Square / StandardVision urban screens:

Work Title	Medium & Tech Stack	Curatorial & Technical Description
`Agua` *(Physical Water + Depth Sensors + Projection)*	Real water basin, Kinect depth sensors, high-contrast video projection, vvvv particle systems (shared engine with *COLAPSO*).	Hybrid physical-digital installation combining real liquid water with depth-tracked generative particle projections. Exploits critical grazing angles and optical refraction on the water's surface to produce an **"infinite mirrorless abyss" effect**, where physical ripples and digital particle physics amplify one another in real time. *(Deployed at scale in Bright Moments Venezia inside a Renaissance church).*
`Fluids` *(2D/2.5D Reactive Shader Relief)*	Custom GLSL/HLSL fragment & compute shaders, real-time 2D fluid velocity maps, normal/displacement texture deformation.	Investigates how purely two-dimensional shader math can evoke visceral three-dimensional materiality and relief. A real-time 2D fluid solver continuously deforms surface textures and specular lighting normals, generating liquid-metal and organic topographic illusions at 60+ fps.
`Colisión` *(Generative 3D Particle Sculptures)*	vvvv, real-time 3D GPU particle collision physics, spatial attractors.	A generative sculpture engine where thousands of 3D bodies undergo continuous physical collision, attraction, and repulsion. Emergent, organic three-dimensional forms crystallize out of controlled chaos, evolving infinitely without ever repeating a frame.
`Rope` *(3D Soft-Body & Flexible Material Simulation)*	vvvv, real-time 3D Verlet integration / constraint physics, PBR material rendering.	An experimental R&D study in real-time 3D soft-body physics, simulating the tension, elasticity, knots, and gravitational sag of flexible ropes and fibrous structures responding to virtual and sensor-driven forces.

3.7 Landmark Site-Specific & Public Art Precedents (`Bright Moments Venezia` & `Coloso de Tecnópolis`)

- **`Bright Moments Venezia & Worldwide` (2023–2024 — Venice, Italy & Buenos Aires):**
  - **Venice (Renaissance Church Site-Specific Installation):** Directed the technical and experiential integration of a real-time fluid simulation, reactive architectural DMX lighting, physical water pooled on the church floor, and generative sound—calibrated remotely prior to load-in using **3D LiDAR scans** of the historic interior.
  - **Buenos Aires (8-Hour Live Generative Performance Marathon):** Engineered a real-time parameter-capture and DMX lighting cue system that translated 200 live-minted generative art pieces into spatial light beams on the fly, directing dancers, theater performers, and musicians across an un-rehearsed 8-hour marathon (Watch Video).
- **`Coloso de Tecnópolis` (2012–2024 — with DOMA Collective, Buenos Aires):**
  - Engineered and maintained the custom real-time video-to-LED control pipeline for *Coloso*, an iconic **45-meter-tall outdoor public sculpture** at Tecnópolis. Long before commercial pixel-mapping servers were commonplace, Palenque and DOMA Collective built a custom software-to-hardware video animation pipeline that kept the monumental structure running and evolving outdoors for over a decade.

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4. Turnkey Commissioning, Grant & Residency Readiness

Manuel Palenque and **DataRaíz** offer end-to-end, turnkey production capabilities for museums, curatorial agencies, public art commissions, architectural developers, and brand innovation labs:

1. **Concept, Spatial Architecture & 3D Digital Twin Pre-Visualization:** Complete CAD/3D stage and gallery modeling, LiDAR scan integration, sightline analysis, and real-time virtual simulation of projections and DMX lighting prior to physical fabrication.
2. **Custom Real-Time Software & Shader Engineering:** Bespoke, fault-tolerant engines built in `vvvv`, `TouchDesigner`, `WebGPU`, `Python`, and custom `GLSL/HLSL` shaders—engineered for 24/7 unattended museum operation or zero-latency live stage performance.
3. **Sensor, Hardware & Lighting Integration:** Multi-camera **Kinect / Orbbec / LiDAR** depth tracking, **DMX512 / Art-Net / sACN** lighting and **ILDA laser** synchronization, multi-projector blending/warping (including 360° fulldome), and distributed **QR / WebSockets** mobile audience participation systems.
4. **International Deployment & US Routing:** Active **US Visa**, bilingual fluency (English / Spanish), and frequent travel corridors through **New York City, Chicago, and Miami** ensure streamlined on-site installation, artist residencies, and keynote engagements across North America.

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5. Direct Contact & Representation Inquiries

- **Artist / Speaker / Creative Technologist:** Manuel Palenque
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- **Main Portfolio:** https://mpalenque.github.io/
- **Syncro A/V Lecture Portal:** https://mpalenque.github.io/syncro-av/
- **Immersive Design & Creative AI Course:** https://superside-immersive.github.io/course/index.html
- **DataRaíz Media Lab & Ciclo Negro:** https://dataraiz.io · https://ciclonegro.dataraiz.io/
- **Press Kit & Archival Drive:** https://drive.google.com/drive/folders/192TdHTU385tIgXai0DBxTlZSCykc6Fdb?usp=sharing